

Practical rules we can use

1. Establish a Clean Baseline

A model should not learn from fraudulent or anomalous data.

Rules:

- Train only on verified legitimate behavior.
- Exclude periods containing known fraud incidents.
- Remove corrupted or incomplete records.
- Filter out system errors and duplicated transactions.
- Use a minimum history threshold.

Example:

Only build a behavior profile if the customer has:

- ≥90 days history
 - ≥100 transactions
 - No confirmed fraud cases
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2. Minimum Data Before Building Behaviour

Never create a profile with insufficient data.

Example thresholds:

Attribute	Minimum
Days active	60–90 days
Transactions	50–500
Login events	100
Devices	3+ observations

Locations 5+
observations

Otherwise:

Profile Status = Warming Up

3. Retraining Frequency

Behavior changes over time.

Typical schedules:

High-risk banking

- Daily

Corporate users

- Weekly

Retail customers

- Weekly or Monthly

Trading systems(for NFT) - i believe we have a customer coming for it

- Near real-time

4. Retraining Only When Enough New Data Exists

Avoid retraining on tiny changes.

Example:

Retrain if:

New Transactions ≥ 100

OR

30 days elapsed

OR

Significant drift detected

5. Sliding Window Learning

Avoid learning from data that's too old.

Instead of:

2019 $\rightarrow$ 2026

Use:

Last 90 days

or

Last 180 days

or

Last 12 months

This keeps profiles relevant.

6. Forgetting Rule (Behavior Decay)

Older behavior should gradually lose influence.

Example:

Today's transaction

Weight = 1.0

30 days old

Weight = 0.8

90 days

Weight = 0.5

180 days

Weight = 0.2

This is known as **time decay**.

7. Never Learn Confirmed Fraud

Once an event is confirmed fraudulent:

- remove it from future training
- remove linked transactions if necessary
- rebuild the profile if it polluted training

8. Behaviour Stability Rule

Don't immediately change a user's profile because of one unusual event.

Example:

Normal login time:

8AM

One login at 2AM

Do NOT change profile.

Require repeated evidence.

Example:

New pattern observed 10 times

THEN

Update behaviour profile

9. Drift Detection

Detect whether the user's behavior has genuinely changed.

Examples:

- moved to another country
- new employer
- changed spending habits
- salary increase
- relocation

If drift is gradual:

Adapt slowly

If drift is sudden:

Flag first

Then monitor

Only learn after validation

10. Confidence Threshold

Every profile should have a confidence score.

Example:

Confidence =

history length

+

consistency

+

data completeness

Example:

Confidence

95%

Profile trusted

45%

Still learning

11. Retrain Only After Verification

If fraud analysts mark alerts:

Confirmed Fraud

Never learn

False Positive

Safe to learn

True Positive

Exclude

Customer Verified

Learn

Human feedback significantly improves model quality.

12. Version Behaviour Profiles

Keep versions of user profiles.

Example:

Profile v1

Jan

Profile v2

Feb

Profile v3

March

If retraining degrades performance:

Rollback

14. Retraining Triggers

Trigger retraining when:

- enough new transactions
- significant behavioral drift
- periodic schedule
- new customer segment
- new fraud patterns
- analyst feedback
- feature changes
- model performance degradation

15. Prevent Model Poisoning

Fraudsters may attempt to "teach" the system that fraudulent behavior is normal.

Mitigations:

- don't immediately learn new behavior
- require repeated legitimate observations
- analyst approval for high-risk profile changes
- exclude suspicious periods from retraining
- use anomaly thresholds before profile updates

16. Behaviour Profile Components

A robust behavioral profile may include:

- Transaction amount distribution
- Spending categories
- Merchant types
- Preferred transaction times
- Day-of-week patterns
- Geographic locations
- Device fingerprints
- IP addresses
- Network provider
- Browser or app version
- Login frequency
- Failed login attempts
- Transaction velocity
- Beneficiary relationships
- Payment channels
- Session duration
- Typical balance ranges
- Salary/payment cycles

Example Workflow for Transaction Monitoring

1. Collect transaction, login, device, location, and customer data.
2. Clean and validate the data, excluding known fraud and corrupt records.
3. Wait until a user has enough history to build a reliable profile.
4. Train an initial behavioral profile using a recent sliding window (e.g., 90–180 days).
5. Score new events against the user's baseline and generate anomaly scores.
6. Route high-risk anomalies to analysts for investigation.
7. Feed investigation outcomes (confirmed fraud, false positive, verified legitimate) back into the training pipeline.
8. Detect concept drift and trigger retraining when thresholds are met.
9. Apply time decay so recent behavior influences the profile more than older behavior.
10. Validate the retrained model against a holdout dataset before deployment.
11. Version the model and behavior profiles so changes are auditable and reversible.
12. Continuously monitor production metrics (precision, recall, false-positive rate, latency, and drift) and retrain only when justified.